

ODESSKOYE, Russian Federation

WMO: 287970

Lat: 54.200N

Lon: 72.967E

Elev: 409

StdP: 14.48

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.4	-24.4	-34.3	0.8	-29.2	-29.3	1.1	-24.3	21.8	12.0	18.7	17.2	3.3	260	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	88.5	65.2	85.0	64.5	81.7	63.4	68.8	81.1	67.3	79.4	65.8	77.3	7.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	93.8	72.7	63.2	88.0	71.3	61.5	82.9	70.6	33.2	81.6	32.0	78.8	30.8	77.4	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.3	17.0	14.7	DB	-32.6	93.5	6.3	3.1	-37.1	95.7	-40.8	97.5	-44.3	99.3	-48.9	101.5	(4)
(5)			WB	-32.7	72.3	6.3	2.0	-37.2	73.7	-40.9	74.8	-44.4	75.9	-49.0	77.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.9	-0.4	3.2	18.1	40.0	55.4	64.9	66.8	63.1	52.3	39.0	19.7	6.7		
	DBStd	26.51	14.02	12.91	11.32	11.30	9.46	8.06	6.76	7.53	8.55	8.98	14.02	14.00		
	HDD50	6932	1561	1311	990	334	54	4	0	2	72	354	909	1341		
	HDD65	10934	2026	1731	1455	751	320	100	62	127	392	805	1359	1806		
	CDD50	1793	0	0	0	34	222	452	521	410	140	14	0	0		
	CDD65	318	0	0	0	0	22	98	119	69	10	0	0	0		
	CDH74	3430	0	0	0	11	360	1077	1127	712	142	1	0	0		
	CDH80	1074	0	0	0	1	92	364	350	230	37	0	0	0		
(14) Wind	WSAvg	6.9	6.7	7.0	7.8	7.9	7.6	6.3	5.6	5.7	6.2	6.9	7.3	7.6		
(15) Precipitation	PrecAvg	13.8	0.7	0.6	0.7	0.9	1.2	1.8	2.4	1.9	0.9	1.0	1.1	0.8		
	PrecMax	19.0	1.6	1.0	1.5	2.6	3.0	3.5	6.9	3.7	2.3	2.5	2.8	1.6		
	PrecMin	8.8	0.0	0.0	0.1	0.1	0.2	0.6	0.3	0.7	0.3	0.1	0.2	0.2		
	PrecStd	3.0	0.4	0.2	0.4	0.6	0.6	0.7	1.6	0.9	0.5	0.6	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.5	33.3	43.5	74.8	88.2	91.6	93.1	92.1	84.7	67.7	45.8	34.1		
		MCWB	31.7	32.0	39.0	55.4	61.3	64.4	67.5	64.9	60.8	50.9	42.2	32.5		
	2%	DB	28.5	30.1	37.6	68.2	82.0	88.1	87.1	86.1	78.0	62.8	42.5	31.9		
		MCWB	27.6	28.9	35.0	51.6	58.6	64.6	66.5	65.0	58.4	49.7	39.8	30.8		
	5%	DB	24.9	26.8	35.3	63.6	77.9	84.5	83.7	81.4	71.8	57.6	39.1	29.3		
		MCWB	24.0	25.8	33.4	49.8	57.4	63.8	66.2	63.7	55.6	47.1	36.6	28.2		
	10%	DB	20.8	22.5	33.4	58.1	73.3	80.6	80.4	77.3	66.8	52.6	35.4	25.2		
		MCWB	19.8	21.5	31.8	47.6	55.3	63.1	65.4	62.5	53.6	45.3	33.5	24.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.6	32.2	39.0	57.1	63.6	69.8	71.1	69.3	63.9	54.8	43.4	33.1		
		MCDB	32.5	33.0	43.6	70.9	80.5	83.9	81.7	80.9	80.9	64.7	45.0	33.7		
	2%	WB	27.5	28.9	35.3	53.6	60.9	67.5	69.3	67.3	60.0	51.6	40.0	30.8		
		MCDB	28.3	30.0	37.0	65.8	77.5	81.8	81.2	79.6	72.8	59.5	42.1	31.7		
	5%	WB	24.0	25.9	33.7	50.8	58.9	65.7	68.0	65.6	57.3	48.4	36.5	28.2		
		MCDB	24.9	26.9	35.2	61.7	73.5	79.2	79.7	77.1	69.1	55.9	38.8	29.4		
	10%	WB	19.8	21.5	31.9	48.1	56.8	64.0	66.6	63.9	54.9	45.6	33.7	24.1		
		MCDB	20.7	22.5	33.4	57.4	70.6	76.9	77.5	74.8	65.1	52.2	35.1	25.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	13.9	14.3	17.5	22.1	20.3	18.7	19.8	19.0	15.3	11.8	13.0		
		MCDBR	15.0	13.2	11.7	26.9	28.1	26.0	24.0	26.3	26.7	23.9	12.2	13.1		
		MCWBR	14.8	13.0	10.3	13.9	11.6	10.2	9.5	10.7	12.2	13.5	10.2	12.8		
		MCDBR	15.3	13.2	11.6	25.0	25.3	21.6	20.2	21.7	24.2	21.0	12.1	13.3		
	5% WB	MCWBR	15.1	13.0	10.5	13.8	12.4	9.8	9.3	10.0	12.1	13.0	10.5	13.1		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.318	0.376	0.406	0.404	0.411	0.437	0.430	0.370	0.351	0.325	0.291		
	taud		2.046	2.017	1.945	2.142	2.221	2.288	2.203	2.190	2.315	2.254	2.117	2.019		
	Ebn at Noon		198	234	245	259	267	266	255	249	251	224	182	165		
	Edh at Noon		23	33	45	43	42	39	42	40	31	26	21	19		
(44) All-Sky Solar Radiation	RadAvg		340	689	1176	1476	1825	1934	1767	1462	1059	554	263	216		
	RadStd		24	27	103	126	155	103	133	108	78	72	34	30		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page